

# Headphone Output Stage (OPA1622): Working Document

Digital Audio Player • Rev 5

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## 1. Summary

| Field              | Value                                                   |
|--------------------|---------------------------------------------------------|
| Manufacturer / MPN | Texas Instruments, OPA1622IDRCR (#1-056), U103          |
| Package            | 10-pin VSON, thermal pad to V-                          |
| Function           | Dual unity-gain follower driving the 3.5 mm jack (J100) |

## 2. Design Requirements

| Requirement      | Value                                                          |
|------------------|----------------------------------------------------------------|
| Load             | Simgot EW300, $28\ \Omega \pm 15\ \%$ ; 16 to 32 $\Omega$ IEMs |
| Maximum output   | 0.6 V RMS (0.85 V peak)                                        |
| Output impedance | $< 1\ \Omega$ , 20 Hz to 20 kHz                                |
| Supplies         | +3V3_ANALOG / -3V3                                             |
| Shutdown         | EN from GPIO34 (EN_SPEAKER)                                    |
| Constraints      | Low noise • Small footprint                                    |

### 3. Pin Configuration

| Pin   | Name        | Connection                                                            |
|-------|-------------|-----------------------------------------------------------------------|
| 1, 5  | +INA, +INB  | SUMM_R, SUMM_L                                                        |
| 10, 6 | -INA, -INB  | R_fb from the jack node; C_f from the op-amp output                   |
| 9, 7  | OUTA, OUTB  | R_iso to RING / TIP                                                   |
| 2     | V+          | +3V3_ANALOG, 0.1 $\mu$ F + 1 $\mu$ F                                  |
| 4     | V-          | -3V3, 0.1 $\mu$ F + 1 $\mu$ F                                         |
| 3     | GND         | GND (EN reference)                                                    |
| 8     | EN          | 54.9 k $\Omega$ from EN_SPEAKER, 50 k $\Omega$ and 0.1 $\mu$ F to GND |
| Pad   | Thermal pad | V- (confirm on the footprint)                                         |

Source: [1], [4]

### 4. External Components

| Ref                | Value                   | Part / Notes                                                         | Qty |
|--------------------|-------------------------|----------------------------------------------------------------------|-----|
| U103               | Dual op-amp             | TI OPA1622IDRCR (#1-056)                                             | 1   |
| J100               | 3.5 mm jack             | Same Sky SJ1-3535NS (#1-111)                                         | 1   |
| D100               | 2-channel ESD           | TI ESD122DMXR, bidirectional, 0.2 pF, $\pm$ 3.6 V I/O range (#1-112) | 1   |
| R_iso (R125, R127) | 4.7 $\Omega$ 0.1 % 0402 | YAGEO RT0402BRE074R7L (#1-113)                                       | 2   |
| R_fb (R117, R134)  | 499 $\Omega$ 0.1 % 0402 | YAGEO RT0402BRD07499RL (#1-105). Jack node $\rightarrow$ -IN         | 2   |
| C_f (C124, C138)   | 1.5 nF COG 1 % 0603     | Murata GCM1885C1H152FA16D (#1-106). Op-amp output $\rightarrow$ -IN  | 2   |
| C_dec (C128, C134) | 0.1 $\mu$ F X7R 0402    | Murata GCM155R71H104KE02D (#1-080)                                   | 2   |
| C_dec (C129, C135) | 1 $\mu$ F 25 V X7R 0603 | Murata GCM188R71E105KA64D (#1-132)                                   | 2   |
| R_EN1 (R128)       | 54.9 k $\Omega$ 0402    | EN_SPEAKER $\rightarrow$ EN (#1-079)                                 | 1   |
| R_EN2 (R130)       | 50 k $\Omega$ 0603      | EN $\rightarrow$ GND (#1-075)                                        | 1   |
| C_EN (C133)        | 0.1 $\mu$ F X7R 0402    | EN $\rightarrow$ GND (#1-080)                                        | 1   |
| R_DET (R118)       | 100 k $\Omega$ 0402     | Detect node $\rightarrow$ +3V3_DIG (#1-131)                          | 1   |
| R_SER (R119)       | 10 k $\Omega$ 0402      | Detect node $\rightarrow$ PLUG_DET (#1-134)                          | 1   |
| R_BLEED (R129)     | 10 k $\Omega$ 0402      | TIP $\rightarrow$ GND (#1-134)                                       | 1   |

Source: [1], [3], [4]

## 5. Equations and Results

$$f_x = \frac{1}{2\pi R_{fb} C_f} = 213 \text{ kHz}; \quad Z_{out}(20 \text{ kHz}) \approx 4.7 \Omega \cdot \frac{499}{|499 - j5305|} = 0.44 \Omega \quad (1)$$

$$V_{EN} = 3.3 \text{ V} \cdot \frac{50}{54.9 + 50} = 1.57 \text{ V} > 0.95 \text{ V}; \quad \tau_{EN} = (54.9 \text{ k}\Omega \parallel 50 \text{ k}\Omega) \cdot 0.1 \mu\text{F} = 2.6 \text{ ms} \quad (2)$$

$$V_{DET} = 3.3 \text{ V} \cdot \frac{10}{110} = 0.30 \text{ V} (\text{unplugged, amplifier off}) \quad (3)$$

Source: [1]

| Load                 | Network                              | Overshoot at 300 pF / 600 pF / 1 nF                        |
|----------------------|--------------------------------------|------------------------------------------------------------|
| 23.8 $\Omega$ C_load | 2 $\Omega$ / 499 $\Omega$ / 1.5 nF   | Clean at all three                                         |
| 10 k $\Omega$ C_load | 4.7 $\Omega$ / 499 $\Omega$ / 1.5 nF | $\approx 10\%$ / 15 to 20% / 30%, settles in 1 to 2 cycles |

## 6. Important Values

| Item                    | Value                                                                          |
|-------------------------|--------------------------------------------------------------------------------|
| Output swing from rail  | 800 mV (no load): $\approx \pm 2.4 \text{ V}$ on $\pm 3.3 \text{ V}$           |
| Common-mode input range | $-1.8 \text{ V}$ to $+2.3 \text{ V}$                                           |
| Gain-bandwidth          | 8 MHz at $G = 1$                                                               |
| Quiescent current       | 2.6 mA per channel; 5 $\mu\text{A}$ per channel in shutdown                    |
| Input voltage noise     | 2.8 nV/ $\sqrt{\text{Hz}}$ at 1 kHz                                            |
| EN thresholds           | High $\geq 0.95 \text{ V}$ max; low $\leq 0.65 \text{ V}$ min over temperature |
| Jack rating             | 1 A, 12 V DC; 5,000 cycles                                                     |

Source: [1], [2]

## 7. Open Problems

| # | Problem                                                                                                               |
|---|-----------------------------------------------------------------------------------------------------------------------|
| 1 | Confirm the OPA1622 footprint's thermal pad is assigned to the V- net (not shown on the symbol). Noted by the author. |

## 8. Revision Log

| Rev    | Date           | Change                                                                                                                                                        |
|--------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 to 4 | 2026-09-26, 27 | Compensation, jack, ESD, plug detect, stability.                                                                                                              |
| 5      | 2026-10-03     | Reviewed against the schematic: decoupling 1 $\mu\text{F}$ X7R 0603 (#1-132); enable net is EN_SPEAKER (GPIO34); designators added. Added: thermal pad check. |

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## 9. Sources

1. OPA1622 Datasheet, SBOS727B (Texas Instruments)
2. SJ1-353XNS Series Datasheet (Same Sky)
3. ESD122 Product Page and Datasheet (Texas Instruments)
4. DAP schematic sheet DAP-Audio-1, headphone output (author, 2026-10-03)